

Demonstration of Tests of Conversion of PM Dot Notation to Parentheses

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SECTION 0. VERIFICATION TESTS (of dot to paren dot icn)

For each proposition is given:

- 1: the PM notation with dots.
 - 2: the notation with parentheses
 - 3: the Polish (with Lukasiewicz symbols) notation
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*2·06†: $p \supset q \supset : q \supset r \supset . p \supset r$

*3·47†: $p \supset r . q \supset s \supset : p . q \supset . r . s$

*4·22†: $p \equiv q . q \equiv r \supset . p \equiv r$

*4·41†: $p \vee q . r \equiv : p \vee q . p \vee r$

*4·43†: $p \equiv : p \vee q . p \vee \sim q$

*4·44†: $p \equiv : p \vee . p . q$

*4·87†: $p . q \supset . r \equiv : p \supset . q \supset r \equiv : q \supset . p \supset r \equiv : q . p \supset . r$

*4·88†: $p . q \supset . r \equiv : p \supset . q \supset r \equiv : q \supset . p \supset r \equiv : q . p \supset . r$

*5·33†: $p . q \supset r \equiv : p : p . q \supset . r$

From Landon D. C. Elkind's Paper in Russell: Vol. 43, no. 1, page 44

*431·441†: $p \vee q \equiv . r \supset s$

*431·442†: $p \vee q \equiv r \supset : s$

*431·443†: $p \vee q \equiv . r \supset : s$

*431·444†: $p : \vee q \equiv r \supset . s$

*431·445†: $p : \vee : q \equiv . r \supset s$

From same, page 54

*431·54†: $p . q : r . s \supset : p . s : r . q$

check longer prop name

Propositions involving quantifiers

$$*9\cdot2\vdash:(x).\psi x.\supset.\psi y$$

$$*9\cdot21\vdash:(x).\psi x\supset\phi x.\supset:(x).\psi x.\supset.(x).\phi x$$

$$*9\cdot22\vdash:(x).\psi x\supset\phi x.\supset:(\exists x).\psi x.\supset.(\exists x).\phi x$$

$$*9\cdot31\vdash:(\exists x).\phi x.\vee.(\exists x).\phi x:\supset.(\exists x).\phi x$$

$$*9\cdot401\vdash:p:\vee:q.\vee.(\exists x).\psi x:\supset:q:\vee:p.\vee.(\exists x).\psi x$$

$$*10\cdot35\vdash:(\exists x).p.\psi x.\equiv:p:(\exists x).\psi x$$

$$*11\cdot2\vdash:(x,y).\phi[x,y].\equiv.(y,x).\phi[x,y]$$

One Step in proof of 11.55 I wanted example of 2 adjacent quantifiers -
hard to find.

$$*11\cdot551\vdash:(x):(\exists y).\psi x.\phi[x,y].\equiv:\psi x:(\exists y).\phi[x,y]$$

From same, page 46

$$*431\cdot46\vdash:(x).\psi x.\phi x.\supset.(x).\psi x$$

Other Tests

$$*99\cdot99\vdash:\sim(\exists x):\sim\psi x.\supset.(x).\sim\psi x$$